B. KOLLAR

Embedded Software Engineer

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EDUCATION

9/2026 – 9/2027 (Expected)
University of Manchester — **MSc Robotics**
Modules covering: Autonomous Mobile Robots, Reinforcement Learning, Cognitive Robotics and Computer Vision, Robotic Manipulators, Software for Robotics.

9/2021 – 6/2024
University of Manchester — **BSc (Hons) Computer Science**
Upper Second-Class Honours (69%)

- **Third year project**: Researching wiring sparsity via the Deep-R algorithm in spiking neural networks for use with Dynamic Vision Sensor datasets & neuromorphic hardware.
  Linear algebra such as vector spaces, tensor spaces, eigendecomposition, SVD, applied to multiple topics such as machine learning, quantum computing & quantum algorithms such as Quantum Fourier Transform, Shor's algorithm.
- **Natural Language Processing**: Text preprocessing, N–gram language models, part-of-speech tagging & chunking, word embeddings (Word2Vec, GloVe), RNNs/LSTMs, attention mechanism & Transformers, Named Entity Recognition with Conditional Random Fields. Course assignment on sentiment classification for the IMDb review database using Word2Vec, LSTM & BERT.
- **Machine Learning** via RNNs & Transformers with natural language processing; mathematical theory of ML such as bias-variance decomposition, generalisation bounds & Rademacher complexity.
- **RTL chip design** & verification with verilog on Xilinx FPGAs; designed a RISC processor, graphical units, & Mandelbrot set renderer. Programming microcontrollers with ARM assembly; developed an interrupt-based alarm clock & supporting OS library.
- **Algorithms**: complexity analysis, graph search (DFS, BFS, A*), Monte-Carlo tree search, Viterbi, dynamic programming, simplex, Miller-Rabin.
- **3D graphics & Computer Vision**: OpenCV for image processing tasks: edge & object detection, Simultaneous Location & Mapping. 3D graphics theory covering topics such as meshes, collisions, shading models & animation.
- **Discrete mathematics** covering theory of automata, sets, proofs, Context-Free Grammars, & its applications to SAT solvers such as the DPLL algorithm.
- **Operating systems**: low-level hardware specifics covered such as memory management, page tables, caching, pipelining, protection rings, kernel types, interrupts & process models.
- **Group projects**: Hex playing AI using Monte-Carlo Tree search & neural network rollout policy (PyTorch). Othello AI employing MiniMax search with Alpha-Beta pruning (Java). MappedOut: an online geographical social media platform for users to browse or pin posts from popular social media platforms on a global map (Flask, FastAPI, SQLite).

9/2018 – 6/2020
The Bluecoat School, Liverpool — **A-Levels**
*Mathematics, Further Mathematics, Physics, Chemistry (A*AAA)*
Competed twice in the British Physics Olympiad, achieving gold on both occasions. EPQ in Investigating effectiveness of different cyber security strategies for protection & mitigation against common attack vectors.

WORK HISTORY

9/2024 – Present
Embedded Software Engineer — **Bytesnap Design, Birmingham, UK**

- Led software migration for a urinalysis instrument from Windows CE to a new main board with Windows 11: ported ~500k SLoC (C++/C#) to Win32 API, upgrading from .NET Framework 3.5 to .NET 8.0, & developed camera microcontroller firmware for the Infineon FX3/CX3 alongside a DirectShow image library using computer vision techniques (edge detection, colour classification) for identifying samples. Refactored motor control code to an optimised C++ library & custom FPGA comms protocol for real-time performance.
- Developed Yocto BSPs for custom boards targeting i.MX processors: writing device trees, U-Boot & Linux kernel drivers, implemented secure & encrypted boot via OP-TEE & SoC secure enclave or CAAM, & FPGA programming.
- Development for an FPGA-based video mux board with an i.MX SoM & STM32 microcontrollers: developed Linux kernel MIPI DSI driver patches & microcontroller firmware for USB HID device emulation for touchscreen input forwarding.
- Implemented a Diagnostics over IP (ISO 13400) C# stack for ECU factory test tooling, reverse-engineered USB HID descriptors to write a C# userspace driver for a software defined radio & GPS device with a Proprietary USB protocol.
- Android BSP setup, patching & application debugging for Variscite i.MX SoM boards, configuring SELinux.
- System & regression testing of in-house EV load-balancing devices over Modbus; wrote Modbus device simulators for integration testing.

PROJECTS

Double Pendulum Simulation **Python, SymPy**
Numerical Solutions of the Ordinary Differential Equations from the Lagrangian of the system & numerical integrators (Euler, Runge-Kutta, Verlet) for real-time rendering.

Hyperloop Society **Electronics & Software Team** **C/C++**
Designing a base station control UI in C++ & Qt to provide control to the pod's electrical systems.
Work on electronics & embedded systems code in C for STM32 microcontrollers, using CAN bus for subsystem communication & controls.

ip7z-rs **Rust, COM, C++**
Rust crate for bindings to the 7-Zip LZMA SDK via COM FFI.

CAC-Launcher **Rust**
TUI app to distribute mod updates via OneDrive / MSGraph drive manifests & connect to game servers.

OpenGL 3D Renderer **C++**
Using OpenGL for rendering 3D objects from procedural mesh generation. Phong shading or Möller–Trumbore RT shader with web / Emscripten support.